

ZHIQI HUANG

FUKUOKA, JAPAN

EMAIL HOMEPAGE SCHOLAR GITHUB

RESEARCH PROFILE

- M.Phil. student at Waseda University working on 3D understanding and geometry-aware generative modeling across digital humans, PBR materials, volumetric CT, and 3D-aware video/world models.
- Seeking RA and 2027 Ph.D. opportunities in graphics, vision, generative AI, 3D priors, simulation-ready generation, controllable digital humans, and efficient multimodal 3D systems.

RESEARCH INTERESTS

- 3D understanding for generative modeling with depth, normal, material, multi-view, and volumetric priors for 3D, video, and world models.
- 3D-aware video/world models that use spatial priors for embodied interaction, robot manipulation, AR/VR, and interactive simulation.
- Simulation-ready and deployable 3D systems: relightable PBR materials, 3DGS avatars, controllable objects, and efficient consumer-GPU generation.

EDUCATION

Waseda University (早稲田大学)

Apr. 2025–Mar. 2027 (expected)

M.Phil. in Information Architecture

Fukuoka, Japan

- English-taught program
- GPA: 3.8 / 4.0
- Research focus: 3D understanding, geometry-aware generative modeling, and world models for embodied interaction

Sun Yat-sen University (中山大学)

2017–2021

B.E. in Software Engineering

Guangzhou, China

- GPA: 3.7 / 4.0

PUBLICATIONS

SIE3D: Single-Image Expressive 3D Avatar Generation via Semantic Embedding and Perceptual Expression Loss 2026

IEEE ICASSP 2026 (Accepted); First Author, Corresponding Author

- Authors: Zhiqi Huang, Dulongkai Cui, Jinglu Hu
- Uses 3D human priors and multimodal fusion of image identity with text semantics for expression-faithful, language-controllable 3DGS avatar generation.
- Links: Project Page, Paper (arXiv), Code

MANUSCRIPTS

Geometry-aware PBR material generation from long text 2026

First Author; ACM MM 2026 (Under Review)

- Generates relightable PBR maps from long text by using geometry and mesh-normal priors to align material semantics with 3D assets.
- Evaluates semantic alignment, reconstruction quality, multi-view consistency, and consumer-grade GPU inference for simulation-ready 3D asset pipelines.

Deform-then-edit lung-nodule CT forecasting 2026

First Author; Manuscript in preparation; Target: AAAI 2027

- Treats longitudinal CT forecasting as a 3D understanding problem, using deform-then-edit 3D forecasting to predict future ROI appearance while preserving background anatomy.

Text-conditioned PBR texture generation with prompt-derived reference conditions 2026

First Author; Manuscript in preparation; Target: AAAI 2027

- Studies a text-to-reference approach for geometry-aware PBR diffusion: maps material prompts directly into internal reference-condition tokens for the PBR generator, connecting language-level material descriptions with mesh geometry priors to generate coherent albedo, metallic, and roughness maps.

INDUSTRY EXPERIENCE

4399 Games 2023–2024

Senior Graphics Engineer

Guangzhou, China

- Led a 3–5 person rendering team and owned the rendering roadmap for *Era of Conquest* on mobile and PC, balancing 3D scene representation, visual fidelity, and runtime performance.
- Drove graphics development for new mobile, PC, and web projects, connecting reusable 3D asset/material pipelines with practical systems for games and simulation.

4399 Games 2021–2023

Graphics Engineer

Guangzhou, China

- Built and optimized the real-time rendering pipeline for *Era of Conquest*, focusing on cross-platform shader optimization, PBR material representation, and mobile performance profiling.

TECHNICAL SKILLS

- Programming & graphics: C++, Python, C#, GLSL/HLSL; Vulkan, OpenGL, Unity, real-time rendering, physically based rendering (PBR).
- ML & 3D research: PyTorch, 3D Gaussian Splatting, diffusion models, CLIP/LongCLIP, mesh processing, PBR material generation, volumetric CT, parameter-efficient fine-tuning, multi-view evaluation.

LANGUAGES & HONORS

- Chinese: Native (Mandarin and Cantonese)
- English: Professional working proficiency (TOEFL iBT: 90)
- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University, 2018–2019